THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 14: **Probability** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A letter is chosen at random from the word 'CALCULATION'. The probability that it is the letter C is: 1

(a) $\frac{1}{11}$ **(b)** $\frac{2}{11}$ **(c)** $\frac{9}{11}$ **(d)** $\frac{3}{11}$

2. A red die and a blue die are rolled together. The probability that the sum is 5 is: 1

(a) $\frac{1}{9}$ **(b)** $\frac{1}{12}$ **(c)** $\frac{5}{36}$ **(d)** $\frac{8}{9}$

3. A coin is tossed three times. The probability of getting at most two heads is: 1

(a) $\frac{3}{4}$ **(b)** 1 **(c)** $\frac{7}{8}$ **(d)** $\frac{1}{8}$

4. 45 cards numbered 1 to 45 are placed in a box, mixed well and one is drawn at random. The probability that the number drawn is not a multiple of 7 is: 1

(a) $\frac{38}{45}$ **(b)** $\frac{8}{9}$ **(c)** $\frac{13}{15}$ **(d)** $\frac{2}{15}$

5. A bag contains 6 red balls and x blue balls. The probability of drawing a blue ball is 4 times the probability of drawing a red ball. The value of x is: 1

(a) 30 **(b)** 24 **(c)** 4 **(d)** 23

6. A bag contains 2 red and 6 blue balls. When x more red balls are added, the probability of drawing a red ball becomes $\frac{1}{2}$. The value of x is: 1

(a) 5 **(b)** 8 **(c)** 4 **(d)** 3

7. A red die and a blue die are rolled together. The probability that neither die shows a 3 is: 1

(a) $\frac{13}{18}$ **(b)** $\frac{25}{36}$ **(c)** $\frac{2}{3}$ **(d)** $\frac{11}{36}$

8. A letter is chosen at random from the word 'TRIANGLE'. The probability that it is a consonant is: 1

(a) $\frac{5}{8}$ **(b)** $\frac{1}{2}$ **(c)** $\frac{3}{8}$ **(d)** $\frac{3}{4}$

9. A red die and a blue die are rolled together. The probability that the sum is 9 is: 1

(a) $\frac{5}{36}$ **(b)** $\frac{1}{9}$ **(c)** $\frac{1}{12}$ **(d)** $\frac{8}{9}$

10. A coin is tossed three times. The probability of getting the same face in the first two tosses is: 1

(a) $\frac{3}{8}$ **(b)** $\frac{5}{8}$ **(c)** $\frac{1}{2}$ **(d)** $\frac{4}{9}$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A bag contains 8 red, 2 blue, 10 green balls. One ball is drawn at random. Find the probability that it is not blue. 2

12. A red die and a blue die are rolled together. Find the probability that the larger number is 4. 2

13. 22 cards numbered 1 to 22 are placed in a box, mixed well and one is drawn at random. Find the probability that the number drawn is a multiple of 3 or 5. 2

14. A bag contains 3 red, 7 blue, 5 green balls. One ball is drawn at random. Find the probability that it is neither blue nor green. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A bag contains 5 red balls and x blue balls. The probability of drawing a blue ball is 3 times the probability of drawing a red ball. Find the value of x. 3

16. A bag contains 6 red and 10 blue balls. When x more red balls are added, the probability of drawing a red ball becomes $\frac{4}{9}$. Find the value of x. 3

17. A red die and a blue die are rolled together. Find the probability that the sum is at least 10. 3

OR

A bag contains 12 red balls and x blue balls. The probability of drawing a blue ball is 2 times the probability of drawing a red ball. Find the value of x. 3

18. A bag contains 8 red and 4 blue balls. When x more red balls are added, the probability of drawing a red ball becomes $\frac{3}{4}$. Find the value of x. 3

OR

A red die and a blue die are rolled together. Find the probability that the sum is less than 5. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Two dice are thrown together. Find the probabilities of these events (in this order): 5
(i) the sum of the numbers is 7,
(ii) the product is 6,
(iii) the sum is 5 (in this order).

OR

A box contains 45 cards numbered 1 to 45. One card is drawn at random. Find the probability that the number is: 5
(i) a number divisible by 5,
(ii) a prime number,
(iii) a multiple of 2 but not of 4 (in this order).

20. Two dice are thrown together. Find the probabilities of these events (in this order): 5
(i) both numbers are odd,
(ii) the product is a perfect square,
(iii) the sum is a multiple of 4 (in this order).

OR

A box contains 30 cards numbered 1 to 30. One card is drawn at random. Find the probability that the number is: 5
(i) a multiple of 3 or 5,
(ii) a perfect square,
(iii) a number less than 10 (in this order).

END OF QUESTION PAPER — **Total: 40 Marks**